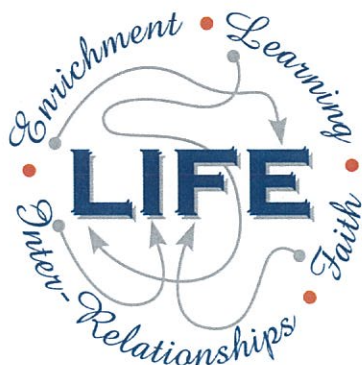




Holy Cross College

Semester 2 Examination 2015

Question/Answer Booklet

**Year 10 Science
(General)**

Please place your student identification label in this box

Student Name

SOLUTIONS

Student's Teacher

Time allowed for this paper

Reading time before commencing work: 10 minutes
Working time for paper: 110 minutes

Materials required/recommended for this paper

To be provided by the supervisor

This Question/Answer Booklet
Data Sheet

To be provided by the candidate

Standard items: pens, pencils, eraser, correction fluid, ruler, highlighters.

Special items: non-programmable calculators satisfying the conditions set by the School Curriculum and Standards Authority for this course.

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Suggested working time (minutes)	Marks available	Percentage of exam
Section One: Multiple Choice	35	35	35	35	25
Section Two: Short Answer	16	16	55	91	63
Section Three: Comprehension	1	9	20	18	12
				144	100

Instructions to candidates

- The rules for the conduct of examinations at Holy Cross College are detailed in the College Examination Policy. Sitting this examination implies that you agree to abide by these rules.
- Write your answers in this Question/Answer Booklet.
- Working or reasoning should be clearly shown when calculating or estimating answers.
- You must be careful to confine your responses to the specific questions asked and to follow any instructions that are specific to a particular question.
- Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.
 - Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
 - Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number.
 - Fill in the number of the question(s) that you are continuing to answer at the top of the page.

SECTION 1 MULTIPLE CHOICE (35 marks)

Choose the best answer for each question in the Multiple Choice Question booklet.

Mark your selection with a cross.

-
- | | | | | | | | | | |
|----|----------------|----------------|----------------|----------------|----|-----|----------------|----------------|----------------|
| 1 | (a) | (b) | (c) | (d) | 31 | (a) | (b) | (c) | (d) |
| 2 | (a) | (b) | (c) | (d) | 32 | (a) | (b) | (c) | (d) |
| 3 | (a) | (b) | (c) | (d) | 33 | (a) | (b) | (c) | (d) |
| 4 | (a) | (b) | (c) | (d) | 34 | (a) | (b) | (c) | (d) |
| 5 | (a) | (b) | (c) | (d) | 35 | (a) | (b) | (c) | (d) |
| 6 | (a) | (b) | (c) | (d) | | | | | |
| 7 | (a) | (b) | (c) | (d) | | | | | |
| 8 | (a) | (b) | (c) | (d) | | | | | |
| 9 | (a) | (b) | (c) | (d) | | | | | |
| 10 | (a) | (b) | (c) | (d) | | | | | |
| 11 | (a) | (b) | (c) | (d) | | | | | |
| 12 | (a) | (b) | (c) | (d) | | | | | |
| 13 | (a) | (b) | (c) | (d) | | | | | |
| 14 | (a) | (b) | (c) | (d) | | | | | |
| 15 | (a) | (b) | (c) | (d) | | | | | |
| 16 | (a) | (b) | (c) | (d) | | | | | |
| 17 | (a) | (b) | (c) | (d) | | | | | |
| 18 | (a) | (b) | (c) | (d) | | | | | |
| 19 | (a) | (b) | (c) | (d) | | | | | |
| 20 | (a) | (b) | (c) | (d) | | | | | |
| 21 | (a) | (b) | (c) | (d) | | | | | |
| 22 | (a) | (b) | (c) | (d) | | | | | |
| 23 | (a) | (b) | (c) | (d) | | | | | |
| 24 | (a) | (b) | (c) | (d) | | | | | |
| 25 | (a) | (b) | (c) | (d) | | | | | |
| 26 | (a) | (b) | (c) | (d) | | | | | |
| 27 | (a) | (b) | (c) | (d) | | | | | |
| 28 | (a) | (b) | (c) | (d) | | | | | |
| 29 | (a) | (b) | (c) | (d) | | | | | |
| 30 | (a) | (b) | (c) | (d) | | | | | |

SECTION 2 SHORT ANSWER (91 marks)

Attempt all questions. Write your answers in the spaces provided.

1. How well a metal conducts electricity is given the symbol K. The bigger the K number, the better the metal conducts electricity. Look at the table below.

Metal	K number
cobalt	15
gold	45
lead	5
silver	60
iron	10

- a. Which metal is the **third best** at conducting electricity? (1 mark)

cobalt (1)

- b. Copper is a better conductor than gold. Copper, however, is not as good as silver. In what **range** would copper's K number be? (1 mark)

45-60 (1)

- c. If lead is the standard, its K number is changed to 'one'. The new K number of cobalt would be 3, while for gold it would be 9. What would the new K numbers for silver and iron be? (2 marks)

silver 12 iron 2 (1 mark each)

2. Read the following newspaper article. It appeared in the Brisbane Courier Mail on April 18 1990.

Use this article to answer the questions below.

Children and pets were being poisoned by wood containing lead-based paints, Queensland University of Technology (QUT) scientists said yesterday.

The QUT forensic science unit supervisor, Dr. David Allen, said: "Small children are in danger because they tend to chew whatever is in their grasp, and the renovation boom has seen handymen in the hundreds sanding old walls."

The Analytical Electron Microscopy Facility Unit at QUT is conducting inexpensive and quicktests to determine if paint samples contain lead.

AQUT forensic researcher, Dr. David Cousens said that home renovators were also the most likely victims as fine paint particles could be inhaled when sanding.

He said that symptoms of minor lead poisoning were not easily detected. "However, lead builds up in the body and in severe cases can result in serious nervous disorders," he said.

"If we were to discover lead in samples the home-owner would be wise to remove the old paint by chemical stripping rather than by sanding or burning the paint off."

- a. Which **two** groups of people are at risk from lead poisoning? (1 mark)

Small children + handymen.

- b. What are you advised to do if you are renovating an old house? (2 marks)

Remove old paint by chemical stripping. (1)

Don't sand or burn the paint off. (1)

- c. Give one reason why lead is a dangerous chemical. (1 mark)

Builds up in the body.

Can cause serious nervous disorders

} either - 1 mark

- d. During the term that you were studying Chemistry, you did an extensive study of one room in your house where chemicals were being stored and/or used. Describe one of the more 'dangerous' chemicals by listing:

i) the name of the chemical, Name (1) (1 mark)

ii) why it is dangerous, (2 marks)

• Include "active ingredient". (1)

• Effect on human body and/or environment. (1).

and iii) how you go about making sure that it doesn't cause harm to other people, pets and the environment. (2 marks)

• Explain how to use safely - gloves? Breathing apparatus? (1).

• Mention what may happen that affects others and environment. (1)

- e. i) There are chemicals that we use constantly in our homes or in some of your work places. Name **five products** that you have studied and used, and give an idea of how these are used. (10 marks)

Name: Name ($\frac{1}{2}$)

Use: • Mention main uses, (1)

• Mention safe way to use. ($\frac{1}{2}$)

Name: _____

Use: _____

Name: _____

Use: _____

Name: _____

Use: _____

Name: _____

Use: _____

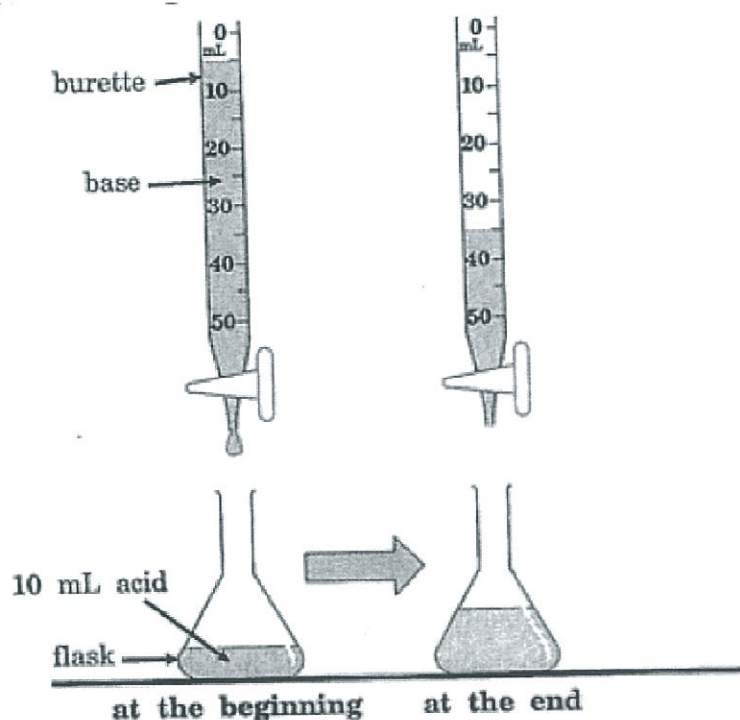
- ii) Why are people still like to use some of these chemicals knowing that they may be harmful? Where do they go to find out more about the dangers of some of these chemicals? (4 marks)

• Still able to do the intended job well. (1)

• Able to take safety precautions to handle them. (1)

• Go to MSDS sheets - supplied by employer or on internet. (2)

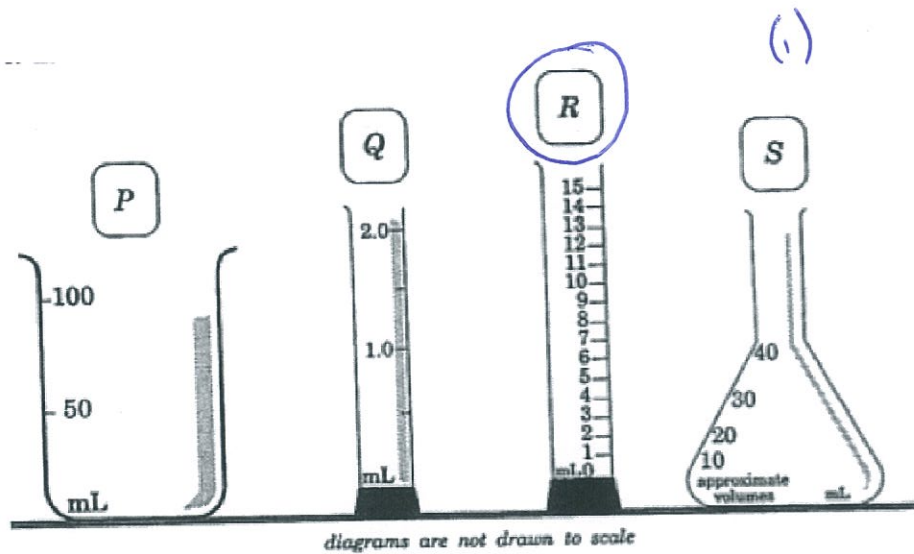
3. A burette is used to accurately add one solution to another. The drawing below shows an experiment using a burette.



- a. How many mL (millilitres) of base was added to the flask from the burette? (1 mark)

30 mL (1)

- b. Which one below would you use to measure the acid accurately before putting it in the flask? Circle or shade in the correct answer. (1 mark)



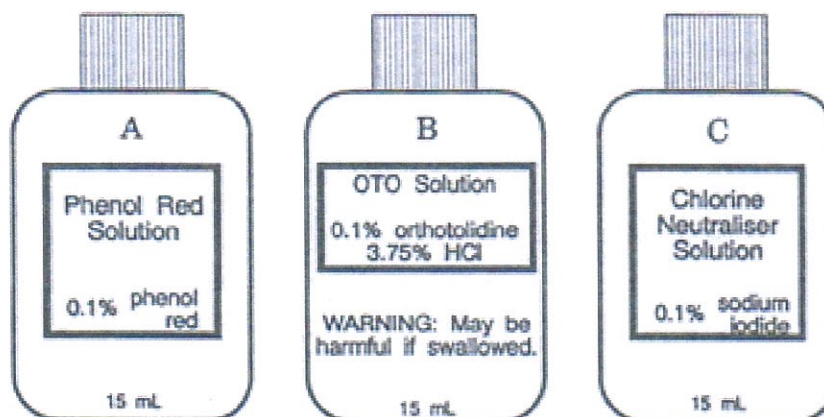
- c. A burette should not be rinsed with water before using it. Suggest a reason why. (2 marks)

• Dilutes the concentration of the solution it holds.

[Mention "dilute" - full marks]

[Mention "changes" - 1 mark only].

4. The instructions for testing pool-water and the chemicals used are given below. The instructions came from the United States of America and therefore use inches (1 in = 2.5 cm).



Chlorine test - For amount of chlorine residual in pool-water 1. Fill the test tube marked 'Cl' with pool-water from 18 inches below the surface in the centre of the pool. 2. Add 5 drops of solution 'B'. 3. Place cap on vial and invert several times to mix the solutions. 4. Compare the colour with the chlorine chart. To obtain a proper chlorine reading, match the colours within 10 seconds. A desirable reading is 1.0 to 1.5. CAUTION: Conduct each test before adding any other pool chemicals. Store the test kit in a cool dark place. For accuracy, use fresh solutions to start each season. Wash hands immediately after testing.	pH test - For degree of acidity or alkalinity of pool water 1. Fill the test tube marked 'pH' with pool-water from 18 inches below the surface. 2. Add 1 drop of solution 'C' and mix it. 3. Add 5 drops of solution 'A'. 4. Place cap on vial and invert several times. 5. Compare the colour with the pH standards. Read the pH level by matching the colours. Between 7.2 and 7.6 is ideal.
--	--

- a. Tom decided to test his pool-water for chlorine (Cl). He rinsed the Cl vial with pool water. He took some water about 40 cm from below the surface of the water near the edge. He added 5 drops of solution 'A' that he bought a couple of years ago. He placed the lid on the vial and shook it a few times. After about 10 minutes, he matched up the colours. He got a reading of 1.9 for the pH.

Tom made **four mistakes** in his method. What are they? (4 marks)

- i) • Took water near the edge.
- Added wrong solution. (A, not B). [1 mark each]
- Waited too long - 10 minutes, not 10 seconds.
- Old solution - not "fresh".

ii) _____

iii) _____

iv) _____

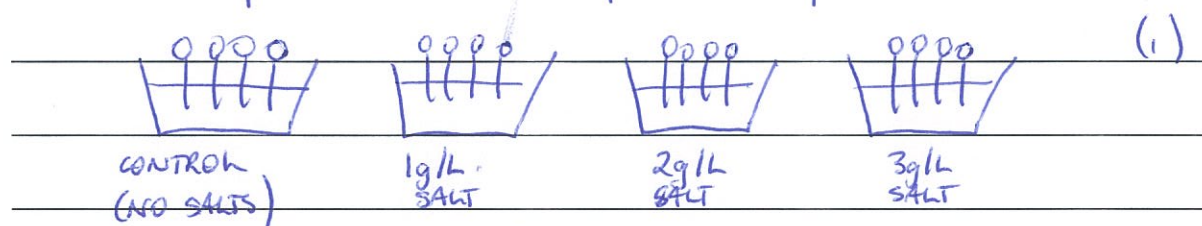
b. State **one thing** you should do when using the solutions safely. (2 marks)

- Conduct each test before adding any other pool chemicals.
 - Store test kit in a dark place.
 - Wash hands after testing.
- } Any one - OK.
- _____

5. Adding certain chemicals to unhealthy plants helps them to grow better. Look at the information below.

Chemical to add	Plants	Treatment
boron	apple, beetroot, cabbage	borax* (10 kg to 30 kg/hectare); keep applying until plants improve
copper	citrus	copper sulfate* (60 to 80 g for young trees, 250 g for older trees); apply to soil every 3 months until trees recover
magnesium	tomatoes	spray with epsom salts; repeat until plants improve
zinc†	citrus stone fruit	spray young trees with zinc sulfate† (15 g/10 L); spray annually in November with zinc sulfate solution (15 g/10 L) Do not repeat either of these treatments unless symptoms recur
NOTES: * Make sure that these salts are in powdered form. Spread over the soil evenly. † Make sure to wash all equipment after use. Zinc is very corrosive.		

- a. What is the maximum amount of borax you would add to **20 hectares** of apples? (1 mark)
- 600 kg
- b. Describe briefly how you would test whether using Epsom salts makes tomatoes grow better. (4 marks)

- Set up "control" and "experimental" plants. (1)
- 
- Place same number of plants, amount of soil, etc in each container. Place all in warm, sunny position. (1)
 - Change concentration of salts. (1)
 - Observe daily over 2-3 weeks, taking photos and recording average height of plants. (1)

- c. What does the information say you should do when adding all of these salts?
(2 marks)

- Make sure they are powder form.
- Spread evenly over the soil.
- Wash all equipment after use

Any 2 - 1 mark each

6. You are asked by your manager at work to trial three different detergents (A, B and C) to see which one will wash and clean the pots, pans and trays you use the quickest and most effectively.

Write up an experiment where you scientifically prove the effectiveness of each of the detergents. You MUST write it up as a scientific report so that he/she can be certain of your decision. Remember that someone else must be able to follow your report, do it themselves using your method, and achieve the same results that you did.

Write it out using the following headings.

(Note: bullet points or short-answer statements may be used.)

AIM:

VARIABLES: independent (changed), dependent (measured) and controlled.

HYPOTHESIS:

MATERIALS REQUIRED:

METHOD USED:

RESULTS: make some up into a table.

CONCLUSION: from your made up results making sure you say exactly why the detergent makes a good choice.

(10 marks)

Aim: To determine which detergent cleans a tray most effectively. (1)

Variables: Independent - Type of detergent
 Dependent - Cleanliness of surfaces. (2)
 Controlled - • Amount of detergent, • Amount of "dirt" - BBQ sauce, • Temperature of water, • Amount of water.

Hypothesis: Detergent B will clean the best. (1)

- Materials:
- BBQ sauce
 - 4x metal trays
 - 4x large plastic containers. (1)
 - Detergents A, B and C.
 - 10 mL measuring cylinder.
 - 25 mL measuring cylinder.

- Method:
1. Prepare 4 trays by spreading 15 mL of BBQ sauce evenly over the surface. Allow it to dry thoroughly. (2)
 2. Place 3.0 L of water at 60°C into each of 4 flat plastic containers.
 3. Add 10 mL of detergent A to container 2, B to container 3 and C to container 4.
 4. Place a tray carefully into each container. Container 1 is the control.
 5. Leave for 5 minutes - then remove carefully.
 6. Take photos of each tray and record the amount of BBQ sauce left (none, a little, a moderate amount, lots).
 7. Repeat for 3 trials.

Results: _____

DETERGENT	AMOUNT OF SAUCE LEFT			
	1	2	3	AVERAGE
CONTROL	lots	lots	lots	lots
A	lots	moderate	moderate	moderate
B	little	none	little	little
C	moderate	little	moderate	moderate

(2)

Conclusion: Detergent B cleans the most effectively. (1)

7. Reaction time is the time taken to react to a stimulus (change in surroundings).

List **3 things** that may affect reaction time and explain how each effect changes the reaction time. (3 marks)

(i) • AGE - improves reaction to 16, then decreases with age.

• ALCOHOL / DRUGS - increases reaction time

(ii)

• DISTRACTIONS - increases reaction time [Any 3 - 3 marks]

(iii) • LEVEL OF CONSCIOUSNESS - fatigue increases reaction time.

TYPE OF STIMULUS - give a variety of reaction times

8. a. Explain how a fatigued (tired) driver could cause an accident on the road. (2 marks)

• Lack of attention - misjudge traffic conditions

• Vehicle wanders on the road.

[Any 2 reasonable causes.]

- b. List **3 ways** you can reduce fatigue (tiredness) as a driver. (3 marks)

(i) • Plan a good sleep before driving.

• Start early in the day.

(ii) • Take regular breaks.

• Share the driving.

[Any 3 - 3 marks]

(iii) • Eat properly.

• Do not have drugs or alcohol.

• Stay overnight on long trips.

9. Explain the difference between **reaction distance** and **braking distance**. (2 marks)

Reaction distance is distance a car moves while the driver reacts to a hazard and moves his/her foot to the brake. (1)

Braking distance is distance covered when the brakes are applied and brings the car to a stop. (1)

10. Explain **three ways** that being under the influence of alcohol can affect your driving ability? (3 marks)

- (i)
 - Slower muscle co-ordination and reflexes.
 - Vision and hearing impaired.
 - Brain's ability to receive and understand messages is reduced.
- (ii)
 - Can't judge speed or distances accurately.
 - Slower decision-making.
 - Increased risk taking - speeding.

(iii) [Any 3 - 3 marks]

11. List **2 reasons** why it is important for people over the age of 80 to re-sit their driver's licence every few years. (2 marks)

- (i)
 - Have slower reflexes and co-ordination.
 - Less acute vision and hearing.
 - Reduced concentration.
- (i)
 - Increased fatigue
 - Less aware of surroundings[Any 2 - 2 marks]

12. What is the difference between a **first** and **second collision** in a car crash.

(2 marks)

First collision is when a car collides with an object or other vehicle. (1)

Second collision is when passengers collide with the inside of the car. (1).

13. (a) What is the difference between a **primary** and **secondary** safety feature?

(2 marks)

Primary safety feature is something that helps prevent an accident.
e.g. Brakes, tyres, vision from the car, steering.

Secondary safety feature is something that helps prevent or reduce injury.

e.g. Seat belts, air bags, crumple zones, protection cells, padded interior.

- (b) Explain why these safety features are effective.

(2 marks)

- Stop passengers moving around in the car during an accident.
- Prevent the vehicle collapsing onto passengers.
- Bring the vehicle/passengers to rest more slowly.

[Any 2 - 2 marks]

14. Read the below table and answer questions below.

SERIOUS CRASHES						
CRASH TYPE	PERTH		COUNTRY		TOTAL	
Rear end	183	14%	32	4%	215	10%
Head on	38	3%	48	6%	86	4%
Side Swipe -same direction	75	6%	28	3%	103	5%
-opposite direction	21	1%	22	3%	43	2%
Right Angle	307	23%	71	9%	378	18%
Turn Across	161	12%	30	4%	191	9%
Hit Pedestrian	181	14%	62	8%	243	11%
Hit Object/Animal	234	18%	267	33%	501	24%
Non-collision	87	6%	232	29%	319	15%
Unknown	37	3%	5	1%	42	2%
Total	1,324	100%	797	100%	2,121	100%

- a. What is the most common form of serious crash? (1 mark)

Hitting an object or animal. (1)

- b. Where do most head on crashed occur? (1 mark)

In the country. (1)

- c. Why would more rear end crashes occur in Perth than in the country? (1 mark)

More vehicles travelling closer together. (1)

- d. Most people blame "the other driver" for accidents. Does the data support this? Explain your answer. (2 marks)

• No. (1)
• 50% of serious crashes do not involve other drivers. (1)

- e. Describe an accident that would be a "non-collision", and explain how a death could occur in this situation. (2 marks)

• Running off a road or rolling the car. (1)
• Passengers are flung violently around inside the car. (1)

15. Sir Isaac Newton discovered the Three Laws of Motion.

- a. Fill in the following definitions of the three laws, **using one word or a phrase to fill in the blank spaces**.

The same word may be used more than once in a definition.

Newton's First Law

Unless a force acts on an object, the object's motion remains the same. (1)

If it is at rest, it stays at rest. (1/2)

If it is moving, it continues to move in a straight (1/2) line at a constant (1/2) speed. (2 marks)

Newton's Second Law

The acceleration of an object is dependent on the size of the force (1/2) acting on it and the mass (1/2) of the object. A larger force (1/2) increases the acceleration of an object, and increasing the mass (1/2) decreases the acceleration of the object. (2 marks)

Newton's Third Law

To every force (reaction) there is an equal and opposite force (reaction). (1 mark)

b. For each of the following situations, state the law of motion that applies and the reason for your choice.

- i) A car can accelerate faster with one person than when it contains five people. (2 marks)

Law 2nd law (1)

Reason Heavier / larger mass $\Rightarrow$ lower acceleration. (1)

- ii) A plasticine figure is placed on a small trolley and released down a ramp. A barricade is placed at the bottom of the ramp. When the trolley hits the barricade, the plasticine figure flies forward. (2 marks)

Law 1st law (1)

Reason Inertia makes the figure continue forwards. (1)

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SECTION 3 COMPREHENSION (18 marks)

Read the article and highlight important words and phrases; then answer the questions that follow. Write your answers in the spaces provided.

Mythconceptions—fry me to the moon

Dr Karl S. Kruszelnicki tackles the myths, curiosities and absurdities of everyday life.

One of the great myths about the space program that got us to the moon was that 'the only good thing to come out of it was Teflon'.

Teflon was in fact discovered, albeit accidentally, much earlier, on April 6, 1938, when a Du Pont chemist, Roy J Plunkett Jr, was looking for a new gas to use in refrigerators. One gas on his list was tetrafluoroethylene. Plunkett opened the valve of what was supposed to be a full cylinder of it, but none came out.

Most people would have just fetched another cylinder, but Plunkett was curious.

He weighed the cylinder and found that it weighed as much as a full cylinder of tetrafluoroethylene, so it wasn't empty. He then proved that the valve was indeed open by running a thin wire

into it. There was only one way left to work out what was going on, so he hacksawed the cylinder open, and found the gas had turned itself into a solid, white, slippery, waxy powder which, when analysed, revealed itself as polytetrafluoroethylene, today known as Teflon.

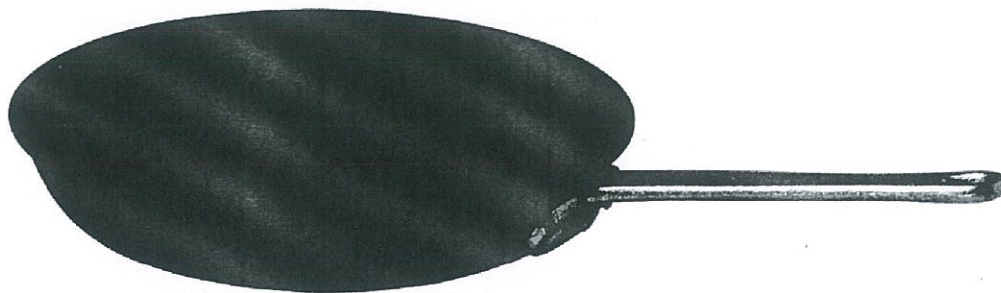
This powder proved to be the most inert plastic Plunkett had ever come across. It was unaffected by low heat. Nothing dissolved it. And it was more slippery than wet ice on wet ice. It was also a terrific electrical insulator. Sadly, it was very expensive to produce, so it had no commercial use.

During World War II, the Manhattan Project saw the first atom bombs made. As part of the manufacturing process, uranium had to be turned into uranium

hexafluoride. Unfortunately, this was an incredibly corrosive gas and would eat through metal, a major problem. But the Manhattan Project had money to burn, and once the US military heard about Teflon, it bought huge quantities of it to protect the machinery from the corrosive uranium hexafluoride.

But Teflon would also have applications in other important areas.

The human immune system tends to ignore Teflon, so it's used in artificial heart valves and aortas, as well as for artificial corneas and bones (for ears, chins, fingers, noses, skulls and various joints). Because it's tough enough to withstand the harsh conditions of outer space, Teflon is also used in spacesuits. And, of course, it's there in frying pans.



1. When was Teflon discovered and by whom? (2 marks)

• April 6 1938 (1)

• Roy Plunkett Jnr (1)

2. What is the correct chemical name of Teflon? (1 mark)

polytetrafluoroethylene (1)

3. Describe what Teflon looks like. (2 marks)

• Solid

• Powder

• White

[Any 4 - $\frac{1}{2}$ mark each]

• Slippery

• Waxy

4. What properties make Teflon an extremely useful material? (2 marks)

• Unaffected by low heat.

• Nothing dissolves it.

• More slippery than wet ice.

[$\frac{1}{2}$ mark each]

• Electrical insulator

5. Teflon is said to be inert - what does this mean? (1 mark)

• Won't react. (1)

6. What research was the inventor doing when he discovered Teflon? (1 mark)

• looking for a new gas to use in refrigeration. (1)

7. What stopped Teflon from being commercially produced in the early days? (1 mark)

• Very expensive to produce. (1)

8. Is Teflon a metal or a plastic? Give one piece of evidence. (2 marks)

• Plastic (1)

• Electrical insulator (1)

9. Complete the following table about the uses of Teflon.

LOCATION	USE	PROPERTY RELATED TO USES
Body	• Heart valves • Aorta • Corneas • Bones	• Immune system ignores it - it's unreactive.
Space	• Space suit	• Tough enough to withstand harsh conditions.
Kitchen	• Frying pans.	• Slippery. • Unreactive.

[1 mark each]

(6 marks)

END OF EXAMINATION